Part A

Answer any one from Part A CO1 10 marks

1. Sales Analysis

A store keeps transaction records in a MySQL table with fields sale_id, product_id, category, quantity, unit_price, and sale_date.

Write code for the following:

Use Sqoop to import the table into HDFS.

Write a Java MapReduce program to calculate total revenue for each product.

Create a Hive table over the imported data and query total revenue by category.

Display the top-selling product and its revenue.

2. Student Marks Summary

A college stores student marks in MySQL with fields student_id, student_name, subject, and marks.

Write code for the following:

Use Sqoop to import the marks table into HDFS.

Use Java MapReduce to calculate the average marks for each subject.

Create a Hive table and query each student's average marks across subjects.

Identify the student with the highest overall average.

3. Library Book Usage

A library maintains a MySQL table with fields transaction_id, student_id, book_id, book_title, issue_date, and return_date.

Write code for the following:

Import the table from MySQL to HDFS using Sqoop.

Write a Java MapReduce program to count the number of times each book was issued.

Create a Hive table and query the number of issues per book title.

Report the most frequently borrowed book.

4. City Sensor Readings

A city stores sensor readings in MySQL with fields reading_id, sensor_id, city, temperature, and reading_time.

Write code for the following:

Import the readings into HDFS using Sqoop.

Use Java MapReduce to calculate the maximum temperature recorded for each city.

Create a Hive table and query the average temperature by city.

List cities where the maximum temperature exceeded a given threshold.

5. Employee Department Analysis

An organization stores employee data in MySQL with fields employee_id, employee_name, department, salary, and joining_year.

Write code for the following:

Use Sqoop to import the employee table into HDFS.

Write a Java MapReduce program to calculate the employee count and total salary for each department.

Create a Hive table and query the average salary by department.

Identify the department with the highest average salary.

Part B

A food delivery platform sends order status updates to Kafka. Each message contains order_id, restaurant_id, delivery_area, status, estimated_delivery_time, cuisine and event_time.

Compute the following:

- Count orders by status and delivery area in one-minute windows.
- Identify delivery areas with a high number of delayed or cancelled orders, using suitably assumed thresholds.
- calculate order counts by cuisine and area in two-minute windows.
- Flag areas where demand exceeds a specified threshold.

CO2 20 marks